

Anand S. Pai

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PROFESSIONAL SUMMARY

Backend engineer with 3+ years of experience building distributed, high-throughput systems in Go. Scaled production APIs from **300 to 1,500+ TPS** with an **~8x latency reduction** (250ms → 30ms). Built fault-tolerant data pipelines processing **20M+ records/day** and migrated **100M+ records with zero downtime**. Experienced in event-driven microservices, gRPC, RabbitMQ, and AWS, with strong knowledge of Kubernetes, Docker and CI/CD. Used **AI-assisted workflows** (**Claude, Cursor**) to accelerate development and code review.

TECHNICAL SKILLS

Languages: Go (Primary), Java, Python, Node.js, TypeScript

APIs & Protocols : REST, gRPC, WebSockets

Architecture: Microservices, Distributed Systems, Concurrency, Fault-Tolerant Design, Event-Driven Architecture

Messaging & Databases: RabbitMQ, Redis, MySQL, OpenSearch, Elasticsearch, AWS DynamoDB

Cloud & DevOps: AWS (S3, EC2, Lambda, DynamoDB), Kubernetes (EKS), CI/CD, Jenkins, Docker

AI & Dev Tools: Cursor, Claude, GitHub Copilot - AI-assisted development, Insomnia, Postman, Draw.io

EXPERIENCE

Tata Communications Ltd. (formerly Kaleyra)

Bengaluru, India

Senior Software Engineer

June 2025 – Present

- Took end-to-end ownership of scaling the **Kaleyra URL Shortener** to handle enterprise-grade traffic, applying horizontal scaling patterns while reducing cloud infrastructure costs.
- Designed backend systems for **OTP delivery and high-priority transactions** with an active-active disaster recovery architecture, ensuring zero-downtime failover under failure scenarios.
- Built a **concurrent token-generation service on AWS Lambda + DynamoDB** achieving **p99 latency ~100ms** with no downtime from collisions or restarts.

Software Engineer

June 2023 – June 2025

- Led migration of **150+ production APIs from PHP monolith to Go microservices using gRPC**, increasing throughput **5x (300 → 1,500+ TPS)** and reducing p95 latency **~8x (250ms → 30ms)**.
- Built a **distributed data export pipeline** processing **20M+ records/day** with batching, concurrency control, and exponential backoff for reliable large-scale delivery.
- Designed a **fault-tolerant Elasticsearch → OpenSearch migration framework** with checkpointing and idempotent processing, migrating **100M+ records with zero downtime**.
- Co-designed **event-driven async pipelines using RabbitMQ and Redis** for SMS campaign workflows, handling millions of message events reliably at scale.
- Implemented secure file upload flow via **ClamAV + AWS S3** and remediated **50+ critical/high VAPT vulnerabilities within one week**, earning commendation from senior leadership.
- Resolved multiple **P1 production incidents** through fast root-cause analysis, consistently meeting SLA commitments with minimal customer disruption.

Kaleyra

Bengaluru, India

Engineer Intern

January 2023 – June 2023

- Built scalable REST APIs for MERN stack projects and developed distributed message-processing workflows using **RabbitMQ** and AWS, contributing to multiple backend services.

EDUCATION

Chandigarh University

Chandigarh, India

B.E. Computer Science – GPA: 8.4 / 10.0

June 2019 – July 2023

PROJECTS

RateLimex : Distributed Rate Limiter – github.com/AnandP201/ratelimex

Java, Spring Boot, Redis, Lua, Docker, Kubernetes, Prometheus, Grafana

- Building a **distributed rate limiting service** capable of handling bursts of traffic

GoStreamMesh : High-Throughput Log Ingestion Platform – github.com/AnandP201/GoStreamMesh

Go, Kafka, Docker, Kubernetes, Prometheus, Grafana

- Actively building a **log ingestion service in Go** using **buffered channels and a configurable worker pool** to be capable of handle production grade traffic

CERTIFICATIONS

Python Data Structures | Using Python to Access Web Data | Solving Problems with JAVA (Honors) – Coursera / UMich

Introduction to Artificial Intelligence – LinkedIn course | Operating System Fundamentals – NPTEL / IIT | Artificial Intelligence and

Machine Learning: Complete Guide (*Pursuing*) - Udemey